EEG During Mental Arithmetic Tasks

objective: To Classify Active state and Inactive state of the signal using raw EEG dataset

Dataset: 16 subjects loaded: each subect correspond to 2 file: before arthmetic task(3 min) raw EEG signals and during arthmetic task raw EEG signals(1min)

Input: 21 features corresponding to 90k rows therefore final dataset size is 90k*21

output: labels 0, 1 active or inactive respectively

3 models were tested in this experiment:

-1) LSTM MODEL -2) BIDIRECTIONAL LSTM -3) LSTM WITH ATTENTION

```
!pip install mne
import mne
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from keras.layers import Conv1D, LSTM , Dense, BatchNormalization,
Input, Bidirectional, Dropout
from keras.models import Model
from sklearn.metrics import classification_report
from sklearn.model_selection import train_test_split
```

```
Collecting mne
  Downloading mne-1.4.2-py3-none-any.whl (7.7 MB)
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 7.7/7.7 MB 29.1 MB/s eta
0:00:00
ent already satisfied: numpy>=1.15.4 in
/usr/local/lib/python3.10/dist-packages (from mne) (1.22.4)
Requirement already satisfied: scipy>=1.6.3 in
/usr/local/lib/python3.10/dist-packages (from mne) (1.10.1)
Requirement already satisfied: matplotlib>=3.4.0 in
/usr/local/lib/python3.10/dist-packages (from mne) (3.7.1)
Requirement already satisfied: tqdm in /usr/local/lib/python3.10/dist-
packages (from mne) (4.65.0)
Requirement already satisfied: pooch>=1.5 in
/usr/local/lib/python3.10/dist-packages (from mne) (1.6.0)
```

```
Requirement already satisfied: decorator in
/usr/local/lib/python3.10/dist-packages (from mne) (4.4.2)
Requirement already satisfied: packaging in
/usr/local/lib/python3.10/dist-packages (from mne) (23.1)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.10/dist-packages (from mne) (3.1.2)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.4.0->mne)
(1.1.0)
Requirement already satisfied: cycler>=0.10 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.4.0->mne)
(0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.4.0->mne)
(4.40.0)
Requirement already satisfied: kiwisolver>=1.0.1 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.4.0->mne)
(1.4.4)
Requirement already satisfied: pillow>=6.2.0 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.4.0->mne)
(8.4.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.4.0->mne)
(3.1.0)
Requirement already satisfied: python-dateutil>=2.7 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.4.0->mne)
(2.8.2)
Requirement already satisfied: appdirs>=1.3.0 in
/usr/local/lib/python3.10/dist-packages (from pooch>=1.5->mne) (1.4.4)
Requirement already satisfied: requests>=2.19.0 in
/usr/local/lib/python3.10/dist-packages (from pooch>=1.5->mne)
(2.27.1)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.10/dist-packages (from jinja2->mne) (2.1.3)
Requirement already satisfied: six>=1.5 in
/usr/local/lib/python3.10/dist-packages (from python-dateutil>=2.7-
>matplotlib>=3.4.0->mne) (1.16.0)
Requirement already satisfied: urllib3<1.27,>=1.21.1 in
/usr/local/lib/python3.10/dist-packages (from requests>=2.19.0-
>pooch>=1.5->mne) (1.26.16)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.10/dist-packages (from requests>=2.19.0-
>pooch>=1.5->mne) (2023.5.7)
Requirement already satisfied: charset-normalizer~=2.0.0 in
/usr/local/lib/python3.10/dist-packages (from requests>=2.19.0-
>pooch>=1.5->mne) (2.0.12)
Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.10/dist-packages (from requests>=2.19.0-
>pooch>=1.5->mne) (3.4)
```

```
Installing collected packages: mne
Successfully installed mne-1.4.2
```

```
from google.colab import drive
drive.mount('/content/gdrive')
```

```
Mounted at /content/gdrive
```

```
subjects= pd.read_csv("/content/gdrive/MyDrive/eeg-during-mental-arithmetic-tasks-1.0.0/subject-info.csv")
subjects
```

```
      Subject  Age Gender  Recording year  Number of subtractions  \
0   Subject00   21      F            2011                    9.70
1   Subject01   18      F            2011                   29.35
2   Subject02   19      F            2012                   12.88
3   Subject03   17      F            2010                   31.00
4   Subject04   17      F            2010                    8.60
5   Subject05   16      F            2010                   20.71
6   Subject06   18      M            2011                    4.35
7   Subject07   18      F            2012                   13.38
8   Subject08   26      M            2011                   18.24
9   Subject09   16      F            2010                    7.00
10  Subject10   17      F            2010                    1.00
11  Subject11   18      F            2010                   26.00
12  Subject12   17      F            2010                   26.36
13  Subject13   24      M            2012                   34.00
14  Subject14   17      F            2010                    9.00
15  Subject15   17      F            2012                   22.18
16  Subject16   17      F            2010                   11.59
17  Subject17   17      F            2010                   28.70
18  Subject18   17      F            2010                   20.00
19  Subject19   22      M            2010                    7.06
20  Subject20   17      F            2010                   15.41
21  Subject21   20      F            2010                    1.00
22  Subject22   19      F            2011                    4.47
23  Subject23   16      F            2010                   27.47
24  Subject24   17      M            2012                   14.76
25  Subject25   17      M            2012                   30.53
26  Subject26   17      F            2010                   13.59
27  Subject27   19      F            2010                   34.59
28  Subject28   19      F            2010                   27.00
29  Subject29   19      M            2011                   16.59
30  Subject30   17      M            2011                   10.00
31  Subject31   19      F            2012                   19.88
32  Subject32   20      F            2011                   13.00
33  Subject33   17      M            2010                   21.47
34  Subject34   18      F            2010                   31.00
35  Subject35   17      F            2010                   12.18
```

```
    Count quality
0                0
1                1
2                1
3                1
4                0
5                1
6                0
7                1
8                1
9                0
10               0
11               1
12               1
13               1
14               0
15               1
16               1
17               1
18               1
19               0
20               1
21               0
22               0
23               1
24               1
25               1
26               1
27               1
28               1
29               1
30               0
31               1
32               1
33               1
34               1
35               1
```

LOADING THE DATASET

THE DATASET FORMAT IS IN MNE FORMAT

SUB_RAW_EEG AND SUB_RAW_EEG STORES THE ARRAY DATA OF EACH SUBJECT AND MNE FORMAT DATA RESPECTIVELY

SIMILARLY SUB_RAW_EEG_AFTER AND SUB_RAW_EEG_AFTER STORE THE ARRAY DATA OF EACH SUBJECT AND MNE FORMAT DATA RESPECTIVELY

```
No_of_subjects = 16
def load_data(n):
    sub_raw_eeg = []
    sub_raw_data =[]
    sub_raw_eeg_after= []
    sub_raw_data_after =[]
    for i in range(n):
        for j in range(2):
            if i<10:
                if j==0:
                    file = "//content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/"+ "Subject0"+ str(i)+"_"+ str(j+1) +
".edf"
                    data_after= mne.io.read_raw_edf(file)
                    raw_data_after = data_after.get_data()
                    sub_raw_data.append(data_after)
                    sub_raw_eeg.append(raw_data_after.T)
                else:
                    file = "//content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/"+ "Subject0"+ str(i)+"_"+ str(j+1) +
".edf"
                    data_after= mne.io.read_raw_edf(file)
                    raw_data_after = data_after.get_data()
                    sub_raw_data_after.append(data_after)
                    sub_raw_eeg_after.append(raw_data_after.T)
            else:
                if j==0:
                    file = "//content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/"+ "Subject"+ str(i)+"_"+ str(j+1) +
".edf"
                    data_after= mne.io.read_raw_edf(file)
                    raw_data_after = data_after.get_data()
                    sub_raw_data.append(data_after)
                    sub_raw_eeg.append(raw_data_after.T)
                else:
                    file = "//content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/"+ "Subject"+ str(i)+"_"+ str(j+1) +
".edf"
                    data_after= mne.io.read_raw_edf(file)
                    raw_data_after = data_after.get_data()
```

```
                    sub_raw_data_after.append(data_after)
                    sub_raw_eeg_after.append(raw_data_after.T)
    return     sub_raw_eeg, sub_raw_data, sub_raw_eeg_after,
sub_raw_data_after


array_list,data,array_after,data_after = load_data(No_of_subjects)
```

```
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject00_1.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject00_2.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject01_1.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject01_2.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject02_1.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject02_2.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject03_1.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject03_2.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject04_1.edf...
```

```
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject04_2.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject05_1.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject05_2.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject06_1.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject06_2.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject07_1.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject07_2.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject08_1.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject08_2.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
```

```
mental-arithmetic-tasks-1.0.0/Subject09_1.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject09_2.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject10_1.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject10_2.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject11_1.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject11_2.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject12_1.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject12_2.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject13_1.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject13_2.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
```

```
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject14_1.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject14_2.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject15_1.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
Extracting EDF parameters from //content/gdrive/MyDrive/eeg-during-
mental-arithmetic-tasks-1.0.0/Subject15_2.edf...
EDF file detected
Setting channel info structure...
Creating raw.info structure...
```

```
data_resampling(data, 170.5,0, 16)
data[3].to_data_frame()
```

```
             time   EEG Fp1   EEG Fp2    EEG F3    EEG F4    EEG F7
EEG F8  \
0        0.000000  7.718681  5.913896  5.423319  8.835119  5.893851
9.877030
1        0.005865  2.263164  1.992290 -0.431128  1.703644 -1.900446 -
0.038821
2        0.011730 -3.354068 -3.203585 -6.982553 -5.169886 -9.527727 -
9.060255
3        0.017595 -3.256600 -5.824088 -8.027228 -5.812735 -9.384481 -
8.165738
4        0.023460  1.233131 -4.976039 -3.623105 -3.909852 -4.149931 -
4.508405
...           ...       ...       ...       ...       ...       ...
...
31026  181.970674  0.000483  0.002728  0.002816  0.005317  0.003290
0.003075
31027  181.976540  0.001508 -0.001654 -0.003246 -0.006626 -0.005739 -
0.003961
31028  181.982405  0.000393  0.002705  0.002820  0.005350  0.003337
0.003079
31029  181.988270  0.001598 -0.001631 -0.003250 -0.006659 -0.005786 -
0.003966
31030  181.994135  0.000302  0.002683  0.002823  0.005383  0.003384
0.003084
```

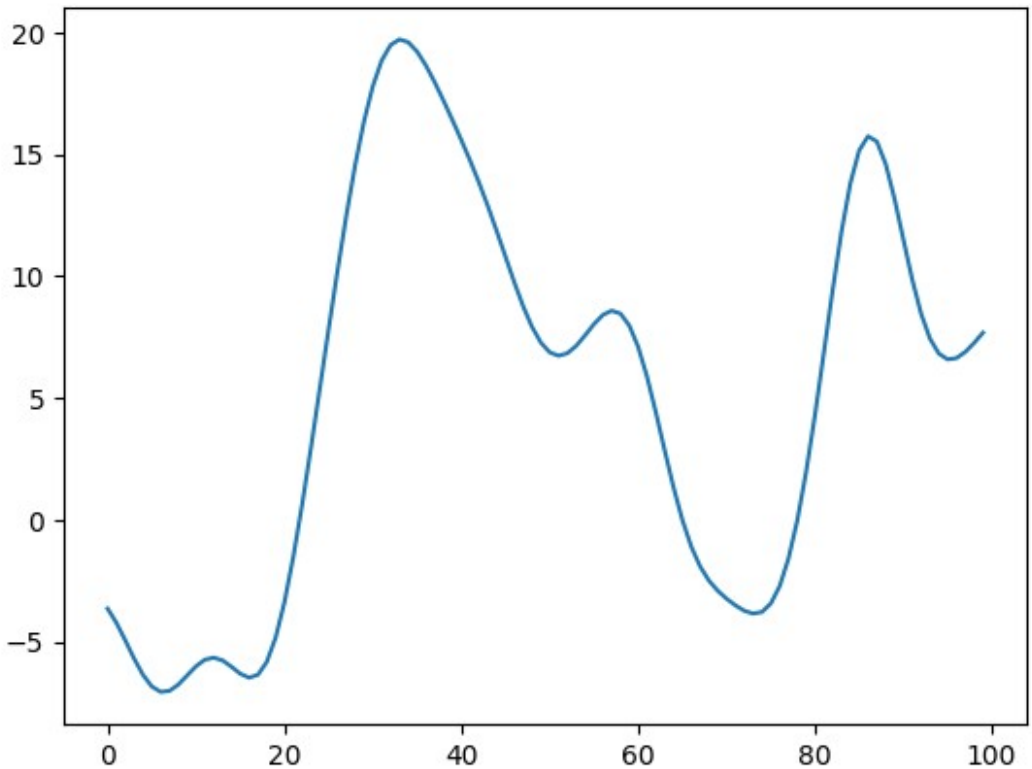
```
         EEG T3    EEG T4     EEG C3  ...     EEG T6     EEG P3
EEG P4  \
0      7.234643  7.732674   5.304332  ...   9.662068  10.919837
4.755852
1     -1.064389  5.597027  -3.190291  ...   5.917179  -0.928378
7.272871
2     -9.124386  5.998133 -10.824619  ...   8.054153 -10.963187
15.195783
3     -9.519064  9.352423 -10.709771  ...  17.111430 -10.579682
24.837203
4     -6.105880  8.237169  -8.549238  ...  16.934400 -11.516365
24.226991
...         ...       ...        ...  ...        ...        ...
...
31026  0.002293  0.001724   0.004667  ...   0.000129   0.002381
0.000433
31027 -0.004942 -0.000112  -0.002969  ...   0.001786   0.001782  -
0.003097
31028  0.002291  0.001705   0.004653  ...   0.000118   0.002348
0.000477
31029 -0.004939 -0.000093  -0.002956  ...   0.001797   0.001815  -
0.003142
31030  0.002288  0.001686   0.004640  ...   0.000107   0.002315
0.000522

         EEG O1     EEG O2    EEG Fz    EEG Cz     EEG Pz  EEG A2-A1
ECG ECG
0      9.598656   5.447699  5.759062  8.119473   4.577504   1.088996
5.805028
1      5.351853   6.893895  0.627751 -1.100233   2.918239  -1.124750
3.331224
2      1.121102  15.756083 -5.230207 -8.335164   6.766877  -1.168037
1.377334
3     -1.978187  29.162477 -6.797721 -7.757907  15.013183   2.007514
1.125199
4     -9.975263  30.004221 -4.431486 -8.862750  14.111482   1.981986 -
6.381725
...         ...        ...       ...       ...        ...        ...
...
31026 -0.000565   0.001974  0.002014  0.002137  -0.001909  -0.000466 -
0.068470
31027  0.002039  -0.002716 -0.001311  0.000809   0.001607  -0.000548
0.074856
31028 -0.000602   0.002040  0.001997  0.002112  -0.001902  -0.000466 -
0.069846
31029  0.002075  -0.002782 -0.001294  0.000834   0.001600  -0.000548
0.076235
31030 -0.000638   0.002107  0.001980  0.002086  -0.001896  -0.000466 -
0.071228
```

```
[31031 rows x 22 columns]
```

```
plt.plot(data[0].to_data_frame()['EEG Fp1'][:100])
```

```
[<matplotlib.lines.Line2D at 0x7f794a9b76a0>]
```




```
data_resampling(data, 170.5, 0, 16)
```

```
data[1].to_data_frame()
```

```
             time   EEG Fp1   EEG Fp2    EEG F3    EEG F4    EEG F7
EEG F8  \
0        0.000000 -6.897114 -7.280397 -5.508518 -3.686438 -6.414296 -
0.027731
1        0.005865 -5.199834 -5.542475 -5.688553 -4.688795 -4.412783 -
2.582951
2        0.011730 -3.002915 -3.306812 -4.423728 -4.698246 -1.921604 -
3.834485
3        0.017595 -2.335808 -2.874836 -2.672722 -2.845106 -1.038197 -
2.741746
4        0.023460 -1.385866 -2.242629 -0.291768  0.956173 -0.378752
1.164667
...           ...       ...       ...       ...       ...       ...
...
31026  181.970674  0.001670  0.001780  0.001239  0.003046  0.000934
```

```
0.009420
31027  181.976540  0.001132 -0.000606  0.001600 -0.000405  0.001380 -
0.004814
31028  181.982405  0.001628  0.001744  0.001231  0.003024  0.000902
0.009298
31029  181.988270  0.001174 -0.000570  0.001609 -0.000384  0.001412 -
0.004692
31030  181.994135  0.001586  0.001707  0.001222  0.003003  0.000871
0.009176

         EEG T3     EEG T4    EEG C3  ...    EEG T6    EEG P3    EEG
P4  \
0     -5.610492  -6.584642 -9.773346  ... -3.109862 -4.933751 -
7.034808
1     -6.855083 -10.933524 -8.927322  ... -6.171873 -5.639834 -
8.765302
2     -4.159783 -12.528650 -6.184393  ... -6.475629 -4.404662 -
8.319983
3      2.290990  -9.218810 -2.057055  ... -2.653116 -0.793643 -
4.681331
4      4.936718  -5.340345  0.760669  ...  0.459201  1.261131 -
1.470610
...         ...        ...       ...  ...       ...       ...       ..
.
31026  0.001309  -0.002563  0.000423  ...  0.002357  0.000246 -
0.001940
31027  0.004807   0.005341  0.000758  ... -0.000371  0.001552
0.000646
31028  0.001242  -0.002588  0.000403  ...  0.002333  0.000178 -
0.001928
31029  0.004874   0.005366  0.000778  ... -0.000347  0.001619
0.000635
31030  0.001176  -0.002614  0.000384  ...  0.002309  0.000111 -
0.001917

         EEG O1    EEG O2    EEG Fz    EEG Cz    EEG Pz  EEG A2-A1
ECG ECG
0     -6.861328 -9.201271 -3.246203 -4.096014 -6.356972  -1.844781 -
17.307610
1     -7.367708 -9.858544 -4.139445 -7.246170 -6.332760   0.583263 -
27.259370
2     -4.358992 -8.045651 -4.765195 -7.679630 -3.740154   2.391465 -
50.562045
3      2.747579 -3.789554 -5.552570 -3.814906  0.310131   1.755711 -
54.976590
4      6.607798 -1.240014 -3.918103  0.444858  0.906166  -0.158779 -
11.024662
...         ...       ...       ...       ...       ...       ...
...
```

```
31026  0.002559  0.002736 -0.000919 -0.003373  0.002624  -0.000991
0.066705
31027 -0.000202  0.000048  0.000269  0.001974  0.001103   0.000607  -
0.059984
31028  0.002447  0.002664 -0.000907 -0.003366  0.002530  -0.000957
0.064837
31029 -0.000090  0.000119  0.000257  0.001968  0.001198   0.000573  -
0.058118
31030  0.002336  0.002592 -0.000894 -0.003360  0.002435  -0.000923
0.062973

[31031 rows x 22 columns]
```

DATA PREPROCESSING

1) DATA RESAMPLLING:- 3 MIN DATA(WHICH IS OF BEFORE ARTHMETIC TASK) IS RESAMPLED INTO THE 1 MIN DATA, THEREFORE 90K ROWS CONVERTED INTO THE 30K

2) DATA LABELLING: RESAMPLED DATA WHIHC IS OF BEFORE ARTHMETIC TASK IS RESAMPLED IS LABELLED 0 AND THE ARTH-MATIC TASK DATA OF 1MIN IS LABELLED 1

3) DATA CONCATATING: BOTH THE DATA OF THE BEFORE ARTHMETIC TASK DATA AND THE AFTER ARTHMETIC TASK DATA IS CONCATED TO FORM THE FINAL DATA THAT IS TO BE FED TO THE MODEL

4) TRAIN_TEST_VALIDATION SPLIT:- USED THE TENSORFLOW DATASET.API FUNCTION TO CREATE THE DATSET. THEN GIVEN THE SPLIT OF THE 70/15/15

```
def data_resampling(data, sfrq, start, stop):
    for i in range(start,stop):
        data[i].resample(sfrq).to_data_frame()


def data_concat(data_before, data_after, start, stop):
    frames = []

    for i in range(start, stop):
        d_b = data_before[i].to_data_frame()
        d_b['labels'] = 0
        d_a = data_after[i].to_data_frame()
        d_a['labels'] = 1
        frames.append(d_b)
        frames.append(d_a)
    model_input_data = pd.concat(frames)

    return model_input_data

def create_dataset(data, data_after, start, stop, sfreq):
```

```
    dataframe = data_resampling(data, sfreq, start,stop)
    dataframe = data_concat(data, data_after, start,stop)
    dataframe.drop('time', axis = 1, inplace = True)
    X_con = dataframe.drop('labels', axis = 1)
    y_con = dataframe['labels']
    y_con = np.array(y_con)
    X_con = np.array(X_con)
    print(X_con.shape, y_con.shape)
    dataset = tf.data.Dataset.from_tensor_slices((X_con,y_con))
    dataset = dataset.shuffle(len(dataset))
    Datset_Size = X_con.shape[0]
    train_size = int(0.85*Datset_Size)
    test_size = int(0.15*Datset_Size)

    train_dataset = dataset.take(train_size)
    test_dataset = dataset.skip(train_size)
    #validation_dataset = remaining_dataset.take(validation_size)
    #test_dataset = remaining_dataset.skip(validation_size)

    train_dataset  = train_dataset.shuffle(len(train_dataset))
    train_dataset = train_dataset.batch(32)


    return train_dataset,test_dataset, X_con, y_con, dataframe
```

```
train_dataset, test_dataset, X_con, y_con, dataframe=
create_dataset(data, data_after, 0, 16, 170.5)
```

```
(991473, 21) (991473,)
```

```
dataframe
```

```
        EEG Fp1   EEG Fp2     EEG F3    EEG F4    EEG F7    EEG F8
EEG T3  \
0     -3.005984 -2.830782  -3.315703  0.190186 -3.207715 -0.719160 -
2.443014
1     -4.909522 -4.979922  -5.712962 -1.153398 -3.892453 -1.517517
0.099585
2     -6.791289 -6.131153  -9.712450 -3.425286 -4.118303 -0.779045
3.996379
3     -6.834537 -5.071248 -11.903124 -4.794655 -4.601263  2.053783
6.590749
4     -5.807639 -4.147401 -10.637694 -5.374189 -4.503584  2.134021
5.924486
...         ...       ...        ...       ...       ...       ...
...
30995  0.000643 -0.001884  -0.000758 -0.000123 -0.001227  0.000123
0.001188
30996  0.000643 -0.001884  -0.000758 -0.000123 -0.001227  0.000123
0.001188
30997  0.000643 -0.001884  -0.000758 -0.000123 -0.001227  0.000123
```

```
0.001188
30998  0.000643 -0.001884  -0.000758 -0.000123 -0.001227  0.000123
0.001188
30999  0.000643 -0.001884  -0.000758 -0.000123 -0.001227  0.000123
0.001188

          EEG T4    EEG C3    EEG C4  ...     EEG P3     EEG P4
EEG O1  \
0       3.242821 -2.525838  2.279258  ...  -0.985222   4.819438
2.206503
1       6.962197 -0.896430  3.791712  ...   5.169935  11.600486
8.597156
2      11.414687  0.117528  4.683077  ...  11.945900  19.319002
17.474888
3      12.949896 -0.127540  3.908124  ...  13.606577  21.495105
23.919082
4       9.579009 -1.320895  0.137801  ...   8.881596  15.191877
22.147000
...          ...       ...       ...  ...        ...        ...
...
30995   0.000178  0.001212  0.000931  ...   0.000223   0.000310
0.000217
30996   0.000178  0.001212  0.000931  ...   0.000223   0.000310
0.000217
30997   0.000178  0.001212  0.000931  ...   0.000223   0.000310
0.000217
30998   0.000178  0.001212  0.000931  ...   0.000223   0.000310
0.000217
30999   0.000178  0.001212  0.000931  ...   0.000223   0.000310
0.000217

          EEG O2    EEG Fz     EEG Cz     EEG Pz  EEG A2-A1    ECG ECG
labels
0       7.884889  0.192554  -3.818521   0.105714   1.642670   9.238842
0
1      14.670477 -1.125483   8.575372   8.175737   0.602282  -2.765945
0
2      22.411214 -3.349668  19.821770  16.488814  -0.233608 -24.926679
0
3      27.189599 -4.716114  15.191842  17.806110   0.561013 -48.397506
0
4      24.962090 -5.071133   3.328170  12.668809   1.271985 -62.045450
0
...          ...       ...        ...        ...        ...        ...
...
30995   0.001252  0.001336   0.000527   0.000425  -0.000367   0.002146
1
30996   0.001252  0.001336   0.000527   0.000425  -0.000367   0.002146
1
```

```
30997   0.001252  0.001336   0.000527   0.000425  -0.000367   0.002146
1
30998   0.001252  0.001336   0.000527   0.000425  -0.000367   0.002146
1
30999   0.001252  0.001336   0.000527   0.000425  -0.000367   0.002146
1

[991473 rows x 22 columns]
```

```
train_dataset_new= train_dataset.take(int(0.825*len(train_dataset)))
validation_dataset = train_dataset.skip(int(0.825*len(train_dataset)))

len(train_dataset_new), len(validation_dataset)
```

```
(21727, 4609)
```

```
shape = X_con.shape[1]
shape
```

```
21
```

LSTM MODEL

1. input shape : (number of features, 1) #(21, 1) as there are 21 columns in our raw eeg dataset and 1 label correspond to it we want to predict

2. layers: - 2 Dense layers - 3 LSTM layers - 1 dense output layer

```
shape = X_con.shape[1]
def create_model(shape):
    ip = Input(shape =(shape,1))
    Dense1 = Dense(64, activation = 'relu')(ip)
    Dense2 = Dense(128, activation = 'relu')(Dense1)
    lstm_1=  LSTM(256, return_sequences = True)(Dense2)
    drop = tf.keras.layers.Dropout(0.2)(lstm_1)
    lstm_3=  LSTM(256)(drop)
    drop2 = tf.keras.layers.Dropout(0.2)(lstm_3)
    Dense_3= Dense(128, activation = 'relu')(drop2)
    outputs = Dense(1, activation='sigmoid')(Dense_3)

    model = tf.keras.Model(ip, outputs)

    model.compile(optimizer='adam',
                  loss="binary_crossentropy", metrics=["acc"])

    return model

model_2 = create_model(shape)

model_2.summary()
```

```
Model: "model"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 input_1 (InputLayer)        [(None, 21, 1)]           0

 dense (Dense)               (None, 21, 64)            128

 dense_1 (Dense)             (None, 21, 128)           8320

 lstm (LSTM)                 (None, 21, 256)           394240

 dropout (Dropout)           (None, 21, 256)           0

 lstm_1 (LSTM)               (None, 256)               525312

 dropout_1 (Dropout)         (None, 256)               0

 dense_2 (Dense)             (None, 128)               32896

 dense_3 (Dense)             (None, 1)                 129

=================================================================
Total params: 961,025
Trainable params: 961,025
Non-trainable params: 0
_________________________________________________________________
```

```
model.fit(X_train, y_train, epochs = 50, verbose=1)
```

```
Epoch 1/50
578/578 [==============================] - 10s 14ms/step - loss:
0.6683 - acc: 0.5711
Epoch 2/50
578/578 [==============================] - 8s 13ms/step - loss: 0.5546
- acc: 0.7081
Epoch 3/50
578/578 [==============================] - 8s 13ms/step - loss: 0.3903
- acc: 0.8268
Epoch 4/50
578/578 [==============================] - 8s 13ms/step - loss: 0.2848
- acc: 0.8866
Epoch 5/50
578/578 [==============================] - 8s 13ms/step - loss: 0.2318
- acc: 0.9107
Epoch 6/50
578/578 [==============================] - 8s 14ms/step - loss: 0.2036
- acc: 0.9214
Epoch 7/50
578/578 [==============================] - 8s 13ms/step - loss: 0.1745
```

```
- acc: 0.9366
Epoch 8/50
578/578 [==============================] - 8s 13ms/step - loss: 0.1749
- acc: 0.9371
Epoch 9/50
578/578 [==============================] - 8s 13ms/step - loss: 0.1450
- acc: 0.9505
Epoch 10/50
578/578 [==============================] - 8s 13ms/step - loss: 0.1336
- acc: 0.9516
Epoch 11/50
578/578 [==============================] - 8s 13ms/step - loss: 0.1194
- acc: 0.9586
Epoch 12/50
578/578 [==============================] - 8s 13ms/step - loss: 0.1212
- acc: 0.9575
Epoch 13/50
578/578 [==============================] - 8s 13ms/step - loss: 0.1049
- acc: 0.9624
Epoch 14/50
578/578 [==============================] - 8s 13ms/step - loss: 0.0976
- acc: 0.9659
Epoch 15/50
578/578 [==============================] - 8s 13ms/step - loss: 0.0896
- acc: 0.9678
Epoch 16/50
578/578 [==============================] - 8s 14ms/step - loss: 0.0754
- acc: 0.9741
Epoch 17/50
578/578 [==============================] - 8s 14ms/step - loss: 0.0760
- acc: 0.9733
Epoch 18/50
578/578 [==============================] - 8s 14ms/step - loss: 0.0710
- acc: 0.9758
Epoch 19/50
578/578 [==============================] - 8s 13ms/step - loss: 0.0910
- acc: 0.9688
Epoch 20/50
578/578 [==============================] - 8s 13ms/step - loss: 0.0593
- acc: 0.9796
Epoch 21/50
578/578 [==============================] - 8s 13ms/step - loss: 0.0538
- acc: 0.9819
Epoch 22/50
578/578 [==============================] - 8s 13ms/step - loss: 0.0577
- acc: 0.9801
Epoch 23/50
578/578 [==============================] - 8s 13ms/step - loss: 0.0439
- acc: 0.9846
```

```
Epoch 24/50
578/578 [==============================] - 8s 13ms/step - loss: 0.0411
- acc: 0.9863
Epoch 25/50
578/578 [==============================] - 8s 13ms/step - loss: 0.0507
- acc: 0.9828
Epoch 26/50
578/578 [==============================] - 8s 14ms/step - loss: 0.0442
- acc: 0.9846
Epoch 27/50
578/578 [==============================] - 8s 14ms/step - loss: 0.0421
- acc: 0.9860
Epoch 28/50
578/578 [==============================] - 8s 14ms/step - loss: 0.0573
- acc: 0.9812
Epoch 29/50
578/578 [==============================] - 8s 14ms/step - loss: 0.0330
- acc: 0.9891
Epoch 30/50
578/578 [==============================] - 8s 14ms/step - loss: 0.0340
- acc: 0.9890
Epoch 31/50
578/578 [==============================] - 8s 14ms/step - loss: 0.0339
- acc: 0.9881
Epoch 32/50
578/578 [==============================] - 8s 14ms/step - loss: 0.0371
- acc: 0.9877
Epoch 33/50
578/578 [==============================] - 8s 14ms/step - loss: 0.0260
- acc: 0.9915
Epoch 34/50
578/578 [==============================] - 8s 14ms/step - loss: 0.0299
- acc: 0.9884
Epoch 35/50
578/578 [==============================] - 8s 14ms/step - loss: 0.0227
- acc: 0.9921
Epoch 36/50
578/578 [==============================] - 9s 15ms/step - loss: 0.0241
- acc: 0.9906
Epoch 37/50
578/578 [==============================] - 9s 15ms/step - loss: 0.0267
- acc: 0.9900
Epoch 38/50
578/578 [==============================] - 8s 15ms/step - loss: 0.0247
- acc: 0.9907
Epoch 39/50
578/578 [==============================] - 8s 14ms/step - loss: 0.0236
- acc: 0.9918
Epoch 40/50
```

```
578/578 [==============================] - 8s 14ms/step - loss: 0.0219
- acc: 0.9920
Epoch 41/50
578/578 [==============================] - 9s 15ms/step - loss: 0.0248
- acc: 0.9915
Epoch 42/50
578/578 [==============================] - 9s 16ms/step - loss: 0.0169
- acc: 0.9934: 0s - loss: 0.0169 - acc: 0.
Epoch 43/50
578/578 [==============================] - 8s 14ms/step - loss: 0.0238
- acc: 0.9915
Epoch 44/50
578/578 [==============================] - 9s 15ms/step - loss: 0.0205
- acc: 0.9927
Epoch 45/50
578/578 [==============================] - 9s 16ms/step - loss: 0.0230
- acc: 0.9929
Epoch 46/50
578/578 [==============================] - 9s 15ms/step - loss: 0.0298
- acc: 0.9905
Epoch 47/50
578/578 [==============================] - 9s 15ms/step - loss: 0.0309
- acc: 0.9882
Epoch 48/50
578/578 [==============================] - 9s 16ms/step - loss: 0.0172
- acc: 0.9942
Epoch 49/50
578/578 [==============================] - 8s 14ms/step - loss: 0.0163
- acc: 0.9949
Epoch 50/50
578/578 [==============================] - 8s 14ms/step - loss: 0.0176
- acc: 0.9939

<tensorflow.python.keras.callbacks.History at 0x2b0c424c0>
```

```
callback = tf.keras.callbacks.EarlyStopping(monitor='loss',
patience=3)
model_2.fit(train_dataset, epochs = 10, callbacks=[callback])
```

```
Epoch 1/10
18300/18300 [==============================] - 725s 39ms/step - loss:
0.6272 - acc: 0.6358
Epoch 2/10
18300/18300 [==============================] - 301s 16ms/step - loss:
0.4991 - acc: 0.7535
Epoch 3/10
18300/18300 [==============================] - 260s 14ms/step - loss:
0.4364 - acc: 0.7929
Epoch 4/10
18300/18300 [==============================] - 268s 15ms/step - loss:
```

```
0.3932 - acc: 0.8191
Epoch 5/10
18300/18300 [==============================] - 276s 15ms/step - loss:
0.3641 - acc: 0.8353
Epoch 6/10
18300/18300 [==============================] - 297s 16ms/step - loss:
0.3448 - acc: 0.8463
Epoch 7/10
18300/18300 [==============================] - 307s 17ms/step - loss:
0.3303 - acc: 0.8541
Epoch 8/10
18300/18300 [==============================] - 312s 17ms/step - loss:
0.3194 - acc: 0.8593
Epoch 9/10
18300/18300 [==============================] - 662s 36ms/step - loss:
0.3125 - acc: 0.8641
Epoch 10/10
18300/18300 [==============================] - 438s 24ms/step - loss:
0.3091 - acc: 0.8658
```

```
<tensorflow.python.keras.callbacks.History at 0x2b08d5e50>
```

```
model_3 = create_model(shape)
model_3.fit(train_dataset_new, epochs = 10,validation_data =
validation_dataset,  callbacks=[callback])
```

```
Epoch 1/10
21388/21388 [==============================] - 320s 15ms/step - loss:
0.6679 - acc: 0.5805 - val_loss: 0.6094 - val_acc: 0.6612
Epoch 2/10
21388/21388 [==============================] - 334s 16ms/step - loss:
0.5962 - acc: 0.6741 - val_loss: 0.5465 - val_acc: 0.7160
Epoch 3/10
21388/21388 [==============================] - 334s 16ms/step - loss:
0.5433 - acc: 0.7187 - val_loss: 0.5060 - val_acc: 0.7470
Epoch 4/10
21388/21388 [==============================] - 333s 16ms/step - loss:
0.5057 - acc: 0.7480 - val_loss: 0.4618 - val_acc: 0.7754
Epoch 5/10
21388/21388 [==============================] - 336s 16ms/step - loss:
0.4756 - acc: 0.7667 - val_loss: 0.4408 - val_acc: 0.7919
Epoch 6/10
21388/21388 [==============================] - 336s 16ms/step - loss:
0.4536 - acc: 0.7824 - val_loss: 0.4174 - val_acc: 0.8042
Epoch 7/10
21388/21388 [==============================] - 355s 17ms/step - loss:
0.4353 - acc: 0.7939 - val_loss: 0.4050 - val_acc: 0.8125
Epoch 8/10
21388/21388 [==============================] - 338s 16ms/step - loss:
0.4207 - acc: 0.8026 - val_loss: 0.3834 - val_acc: 0.8249
```

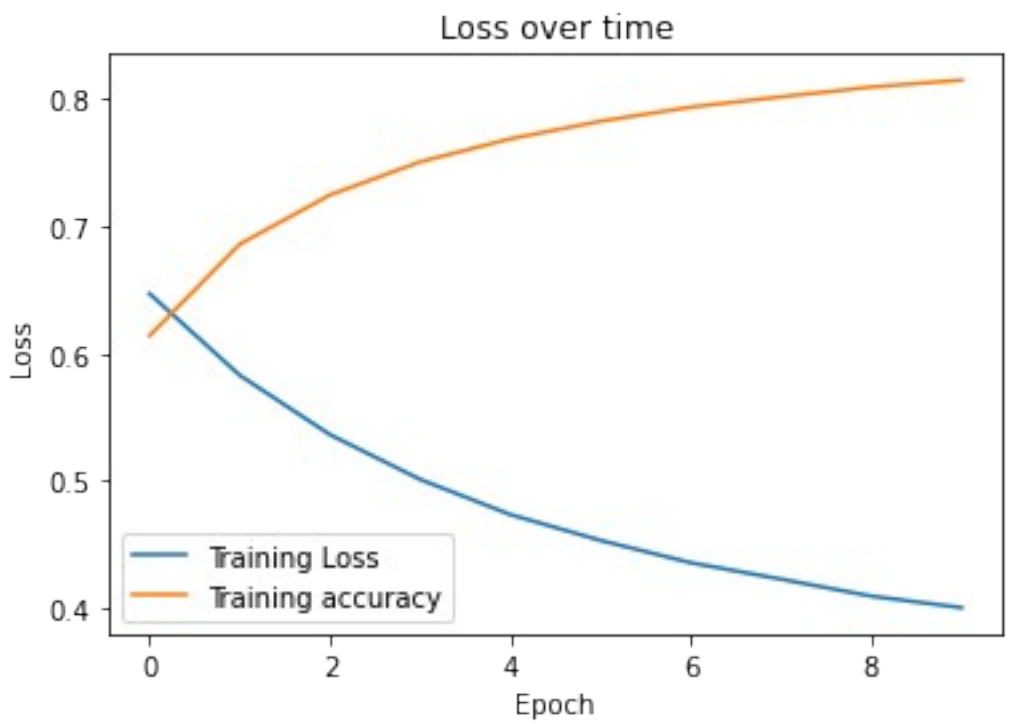
```
Epoch 9/10
21388/21388 [==============================] - 348s 16ms/step - loss:
0.4070 - acc: 0.8105 - val_loss: 0.3738 - val_acc: 0.8306
Epoch 10/10
21388/21388 [==============================] - 342s 16ms/step - loss:
0.3975 - acc: 0.8169 - val_loss: 0.3662 - val_acc: 0.8349
```

```
<tensorflow.python.keras.callbacks.History at 0x322bb7400>
```

```
model_3.save_weights("/content/gdrive/MyDrive/eeg-during-mental-
arithmetic-tasks-1.0.0.h5")
```

```
plt.plot(range(10), model_3.history.history['loss'], label="Training
Loss")
plt.plot(range(10), model_3.history.history['acc'], label="Training
accuracy")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.title("Loss over time")
plt.legend()
```

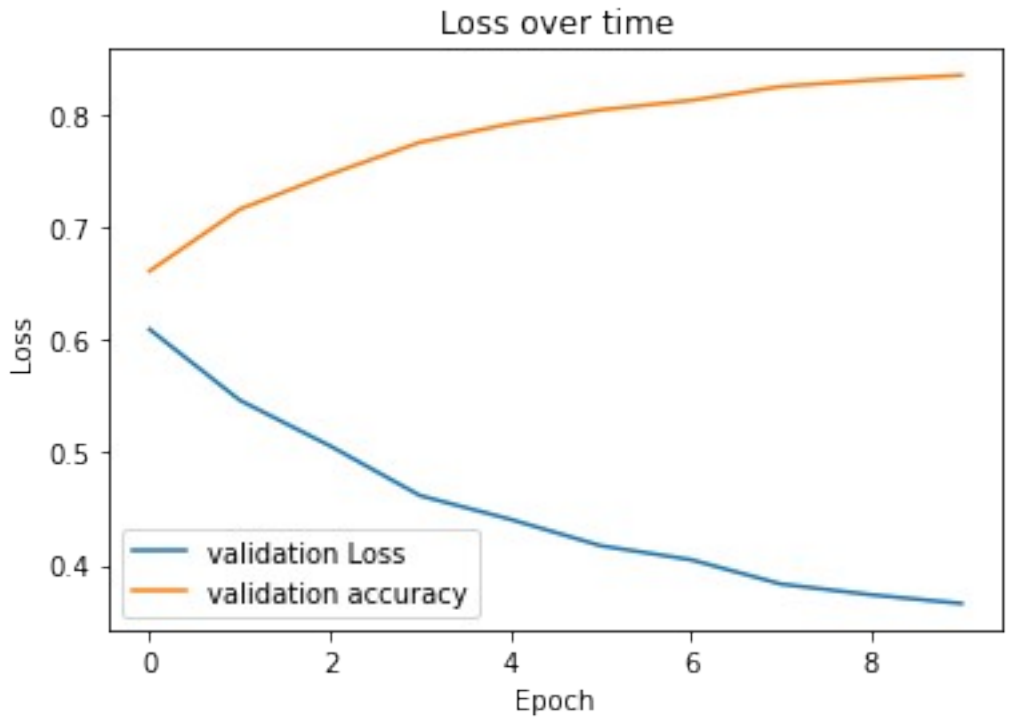
```
<matplotlib.legend.Legend at 0x2faab72b0>
```




```
plt.plot(range(10), model_3.history.history['val_loss'],
label="validation Loss")
plt.plot(range(10), model_3.history.history['val_acc'],
label="validation accuracy")
```

```
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.title("Loss over time")
plt.legend()
```

```
<matplotlib.legend.Legend at 0x2d4515880>
```



Bidirectional LSTM model

```
shape = X_con.shape[1]
def create_model_bilstm(shape):
    ip = Input(shape =(shape,1))
    Dense1 = Dense(64, activation = 'relu')(ip)
    Dense2 = Dense(128, activation = 'relu')(Dense1)
    lstm_1=  Bidirectional(LSTM(256, return_sequences = True))(Dense2)
    drop = tf.keras.layers.Dropout(0.2)(lstm_1)
    lstm_3=  Bidirectional(LSTM(256))(drop)
    drop2 = tf.keras.layers.Dropout(0.2)(lstm_3)
    Dense_3= Dense(128, activation = 'relu')(drop2)
    outputs = Dense(1, activation='sigmoid')(Dense_3)

    model = tf.keras.Model(ip, outputs)

    model.compile(optimizer='adam',
                  loss="binary_crossentropy", metrics=["acc"])

    return model
```

```
model_5 = create_model_bilstm(shape)

model_5.summary()
```

```
Model: "model_42"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
input_51 (InputLayer)        [(None, 21, 1)]           0
_________________________________________________________________
dense_150 (Dense)            (None, 21, 64)            128
_________________________________________________________________
dense_151 (Dense)            (None, 21, 128)           8320
_________________________________________________________________
bidirectional (Bidirectional (None, 21, 512)           788480
_________________________________________________________________
dropout_50 (Dropout)         (None, 21, 512)           0
_________________________________________________________________
bidirectional_1 (Bidirection (None, 512)               1574912
_________________________________________________________________
dropout_51 (Dropout)         (None, 512)               0
_________________________________________________________________
dense_152 (Dense)            (None, 128)               65664
_________________________________________________________________
dense_153 (Dense)            (None, 1)                 129
=================================================================
Total params: 2,437,633
Trainable params: 2,437,633
Non-trainable params: 0
_________________________________________________________________
```

```
model_5.fit(train_dataset_new, epochs = 10,validation_data =
validation_dataset,  callbacks=[callback])
```

```
Epoch 1/10
21727/21727 [==============================] - 1323s 60ms/step - loss:
0.6579 - acc: 0.5991 - val_loss: 0.5875 - val_acc: 0.6816
Epoch 2/10
21727/21727 [==============================] - 1621s 74ms/step - loss:
0.5827 - acc: 0.6861 - val_loss: 0.5383 - val_acc: 0.7208
Epoch 3/10
21727/21727 [==============================] - 927s 43ms/step - loss:
0.5436 - acc: 0.7176 - val_loss: 0.5106 - val_acc: 0.7439
Epoch 4/10
21727/21727 [==============================] - 914s 42ms/step - loss:
0.5200 - acc: 0.7361 - val_loss: 0.4893 - val_acc: 0.7570
Epoch 5/10
21727/21727 [==============================] - 1013s 46ms/step - loss:
0.5037 - acc: 0.7471 - val_loss: 0.4757 - val_acc: 0.7643
Epoch 6/10
```

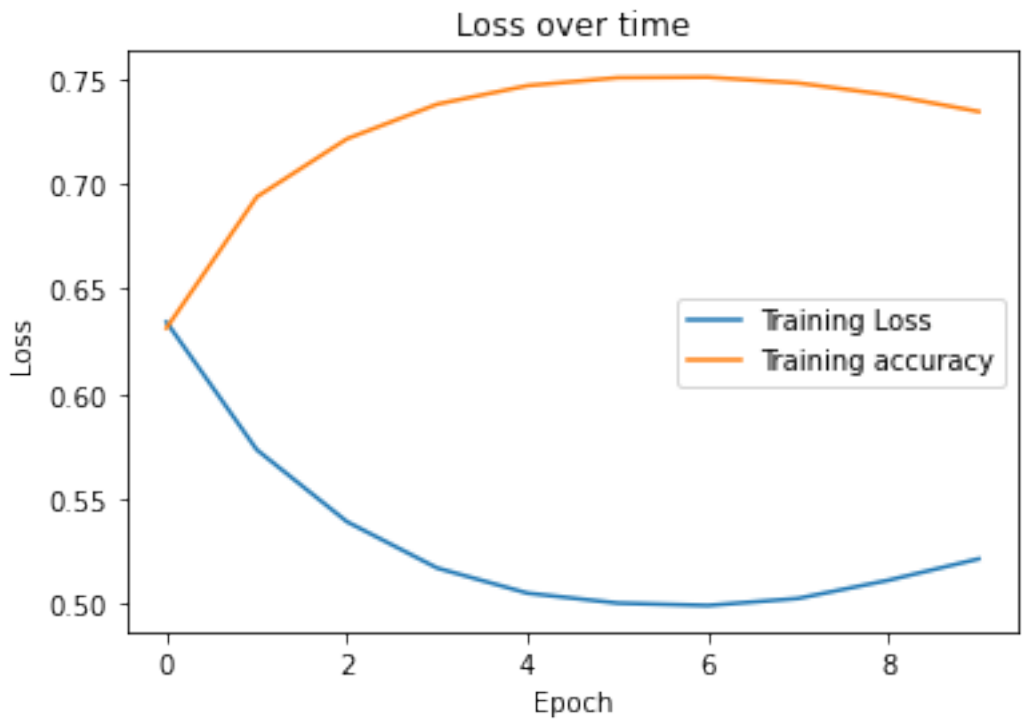
```
21727/21727 [==============================] - 1040s 48ms/step - loss:
0.4977 - acc: 0.7523 - val_loss: 0.4715 - val_acc: 0.7708
Epoch 7/10
21727/21727 [==============================] - 967s 41ms/step - loss:
0.4965 - acc: 0.7528 - val_loss: 0.4752 - val_acc: 0.7661
Epoch 8/10
21727/21727 [==============================] - 912s 42ms/step - loss:
0.4978 - acc: 0.7514 - val_loss: 0.4852 - val_acc: 0.7596
Epoch 9/10
21727/21727 [==============================] - 1381s 63ms/step - loss:
0.5067 - acc: 0.7453 - val_loss: 0.4949 - val_acc: 0.7532
Epoch 10/10
21727/21727 [==============================] - 945s 43ms/step - loss:
0.5168 - acc: 0.7383 - val_loss: 0.5056 - val_acc: 0.7464
```

```
<tensorflow.python.keras.callbacks.History at 0x3c6caff40>
```

```
model_5.save_weights("/content/gdrive/MyDrive/eeg-during-mental-
arithmetic-tasks-1.0.0/model_bilstm.h5")
```

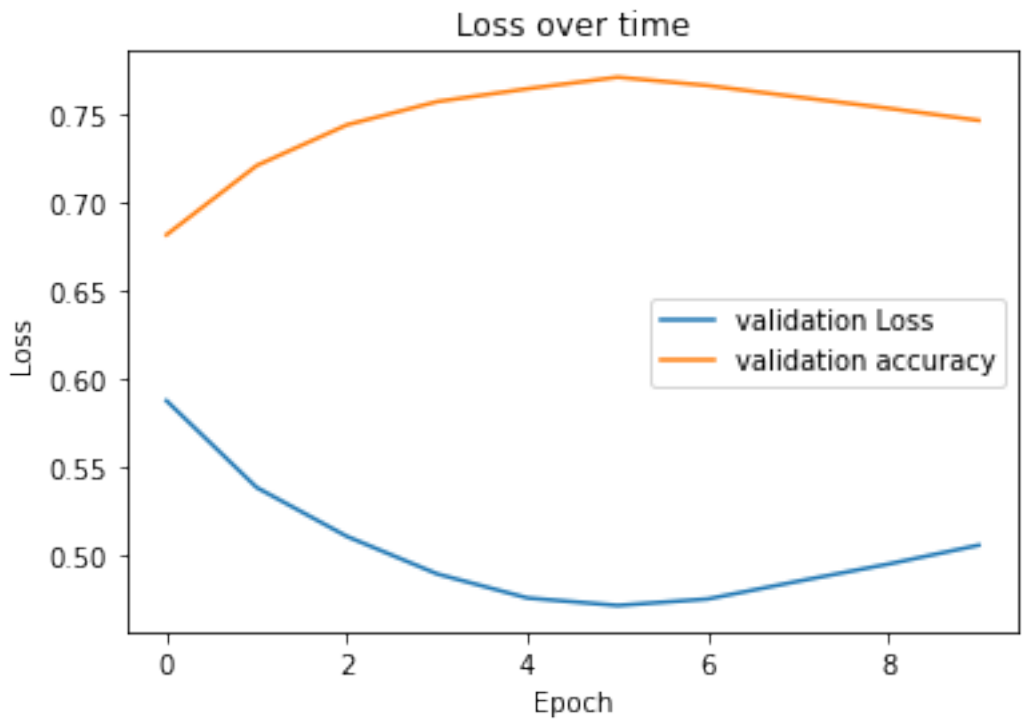
```
plt.plot(range(10), model_5.history.history['loss'], label="Training
Loss")
plt.plot(range(10), model_5.history.history['acc'], label="Training
accuracy")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.title("Loss over time")
plt.legend()
```

```
<matplotlib.legend.Legend at 0x3c72b2190>
```




```
plt.plot(range(10), model_5.history.history['val_loss'],
label="validation Loss")
plt.plot(range(10), model_5.history.history['val_acc'],
label="validation accuracy")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.title("Loss over time")
plt.legend()
```

```
<matplotlib.legend.Legend at 0x440754160>
```



LSTM WITH ATTENTION

```
class Attention_(tf.keras.layers.Layer):

    def __init__(self, units=256, **kwargs):
        self.units = units
        super().__init__(**kwargs)

    def __call__(self, inputs):
        """
        Many-to-one attention mechanism for Keras.
        @param inputs: 3D tensor with shape (batch_size, time_steps,
input_dim this is our case is 256).
        @author: felixhao28, philipperemy.
        """
        hidden_states = inputs
        hidden_size = int(hidden_states.shape[2])
        # Inside dense layer
        #              hidden_states            dot               W
=>           score_first_part
        # W is the trainable weight matrix of attention Luong's
multiplicative style score
        score_first_part = Dense(hidden_size, use_bias=False,
name='attention_score_vec')(hidden_states)
        #            score_first_part           dot
last_hidden_state     => attention_weights
        h_t = Lambda(lambda x: x[:, -1, :],
```

```
output_shape=(hidden_size,), name='last_hidden_state')(hidden_states)
        score = Dot(axes=[1, 2], name='attention_score')([h_t,
score_first_part])
        attention_weights = Activation('softmax',
name='attention_weight')(score)
        # (batch_size, time_steps, hidden_size) dot (batch_size,
time_steps) => (batch_size, hidden_size)
        context_vector = Dot(axes=[1, 1], name='context_vector')
([hidden_states, attention_weights])
        pre_activation = Concatenate(name='attention_output')
([context_vector, h_t])
        attention_vector = Dense(self.units, use_bias=False,
activation='tanh', name='attention_vector')(pre_activation)
        return attention_vector


from tensorflow.keras.layers import Lambda, Dot, Concatenate,
Activation

shape = X_con.shape[1]
def create_model_attention(shape):
    ip = Input(shape =(shape,1))
    Dense1 = Dense(64, activation = 'relu')(ip)
    Dense2 = Dense(128, activation = 'relu')(Dense1)
    lstm_1=  LSTM(256, return_sequences = True)(Dense2)
    drop = Dropout(0.2)(lstm_1)
    lstm_3=  LSTM(256, return_sequences = True)(drop)
    drop2 = Dropout(0.2)(lstm_3)
    att_ = Attention_(256)(drop2)
    #att_ = Attention()(lstm_3)
    Dense_1 = Dense(128, activation = 'relu')(att_)
    outputs = Dense(1, activation='sigmoid')(Dense_1)
    model = tf.keras.Model(ip, outputs)

    model.compile(optimizer='adam',
                  loss="binary_crossentropy", metrics=["acc"])

    return model

model_atten = create_model_attention(shape)

model_atten.summary()
```

```
Model: "model_43"
__________________________________________________________________________________________________
Layer (type)                    Output Shape         Param #     Connected to
==================================================================================================
input_56 (InputLayer)           [(None, 21, 1)]      0
```

```
__________________________________________________________________________________________________
dense_161 (Dense)               (None, 21, 64)       128
input_56[0][0]
__________________________________________________________________________________________________
dense_162 (Dense)               (None, 21, 128)      8320
dense_161[0][0]
__________________________________________________________________________________________________
lstm_114 (LSTM)                 (None, 21, 256)      394240
dense_162[0][0]
__________________________________________________________________________________________________
dropout_56 (Dropout)            (None, 21, 256)      0
lstm_114[0][0]
__________________________________________________________________________________________________
lstm_115 (LSTM)                 (None, 21, 256)      525312
dropout_56[0][0]
__________________________________________________________________________________________________
dropout_57 (Dropout)            (None, 21, 256)      0
lstm_115[0][0]
__________________________________________________________________________________________________
last_hidden_state (Lambda)      (None, 256)          0
dropout_57[0][0]
__________________________________________________________________________________________________
attention_score_vec (Dense)     (None, 21, 256)      65536
dropout_57[0][0]
__________________________________________________________________________________________________
attention_score (Dot)           (None, 21)           0
last_hidden_state[0][0]

attention_score_vec[0][0]
__________________________________________________________________________________________________
attention_weight (Activation)   (None, 21)           0
attention_score[0][0]
__________________________________________________________________________________________________
context_vector (Dot)            (None, 256)          0
dropout_57[0][0]

attention_weight[0][0]
__________________________________________________________________________________________________
```

```
__________________________________________________
attention_output (Concatenate)  (None, 512)          0
context_vector[0][0]

last_hidden_state[0][0]
__________________________________________________
__________________________________________________
attention_vector (Dense)        (None, 256)          131072
attention_output[0][0]
__________________________________________________
__________________________________________________
dense_163 (Dense)               (None, 128)          32896
attention_vector[0][0]
__________________________________________________
__________________________________________________
dense_164 (Dense)               (None, 1)            129
dense_163[0][0]
==================================================================================================
Total params: 1,157,633
Trainable params: 1,157,633
Non-trainable params: 0
__________________________________________________
__________________________________________________
```

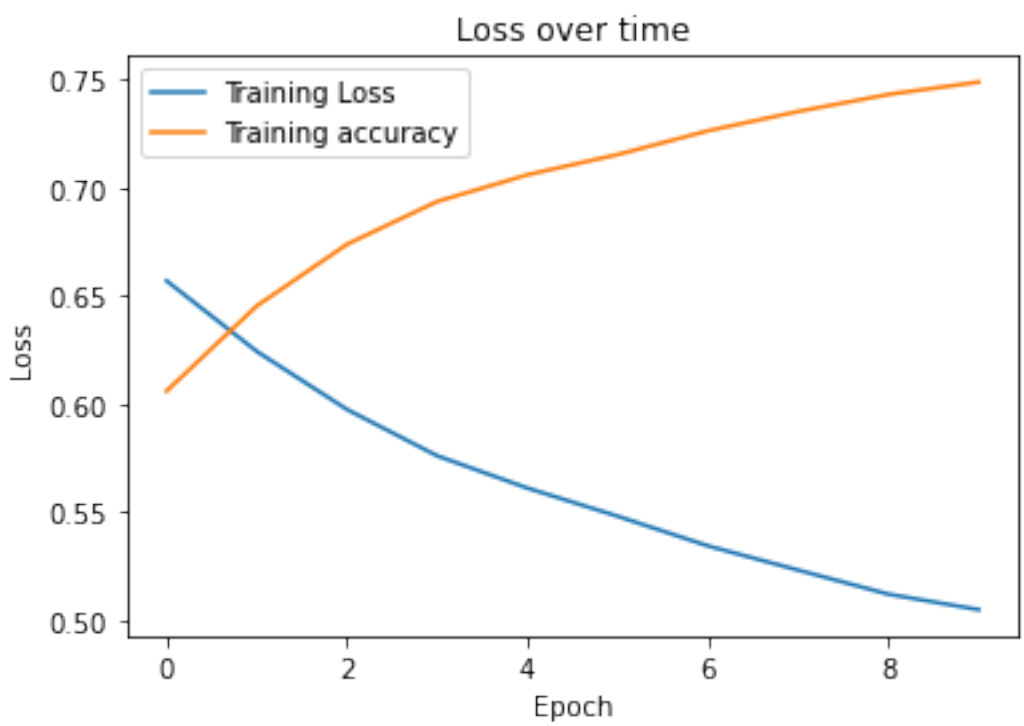
```
model_atten.fit(train_dataset_new, epochs = 10,validation_data =
validation_dataset,  callbacks=[callback])
model_atten.save_weights("/content/gdrive/MyDrive/eeg-during-mental-
arithmetic-tasks-1.0.0/model_atten.h5")
```

```
Epoch 1/10
21727/21727 [==============================] - 1289s 59ms/step - loss:
0.6569 - acc: 0.6059 - val_loss: 0.6363 - val_acc: 0.6333
Epoch 2/10
21727/21727 [==============================] - 426s 19ms/step - loss:
0.6243 - acc: 0.6453 - val_loss: 0.6149 - val_acc: 0.6488
Epoch 3/10
21727/21727 [==============================] - 738s 34ms/step - loss:
0.5974 - acc: 0.6738 - val_loss: 0.5766 - val_acc: 0.6918
Epoch 4/10
21727/21727 [==============================] - 1109s 51ms/step - loss:
0.5760 - acc: 0.6935 - val_loss: 0.5571 - val_acc: 0.7051
Epoch 5/10
21727/21727 [==============================] - 2528s 116ms/step -
loss: 0.5611 - acc: 0.7058 - val_loss: 0.5457 - val_acc: 0.7140
Epoch 6/10
21727/21727 [==============================] - 2538s 117ms/step -
loss: 0.5481 - acc: 0.7152 - val_loss: 0.5305 - val_acc: 0.7286
Epoch 7/10
21727/21727 [==============================] - 1524s 70ms/step - loss:
```

```
0.5344 - acc: 0.7262 - val_loss: 0.5209 - val_acc: 0.7397
Epoch 8/10
21727/21727 [==============================] - 681s 31ms/step - loss:
0.5232 - acc: 0.7351 - val_loss: 0.5040 - val_acc: 0.7481
Epoch 9/10
21727/21727 [==============================] - 883s 41ms/step - loss:
0.5121 - acc: 0.7430 - val_loss: 0.5062 - val_acc: 0.7510
Epoch 10/10
21727/21727 [==============================] - 867s 40ms/step - loss:
0.5050 - acc: 0.7487 - val_loss: 0.4855 - val_acc: 0.7622
```

```
plt.plot(range(10), model_atten.history.history['loss'],
label="Training Loss")
plt.plot(range(10), model_atten.history.history['acc'],
label="Training accuracy")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.title("Loss over time")
plt.legend()
```

```
<matplotlib.legend.Legend at 0x3c735cc40>
```

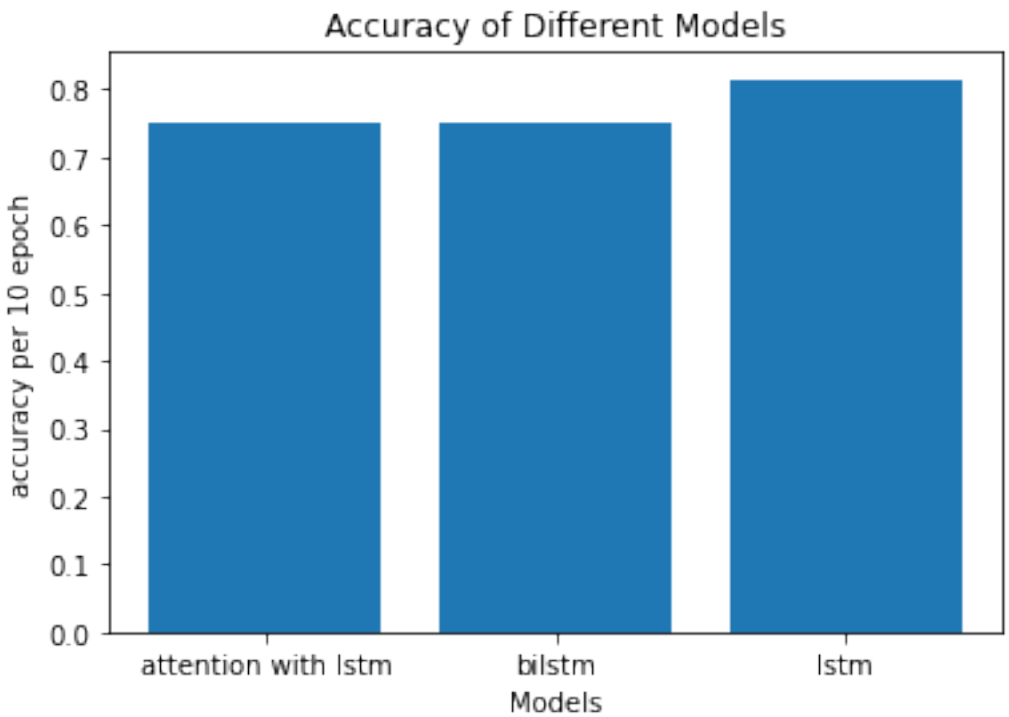


VISUALIZING AND COMPARING THE 3 MODELS

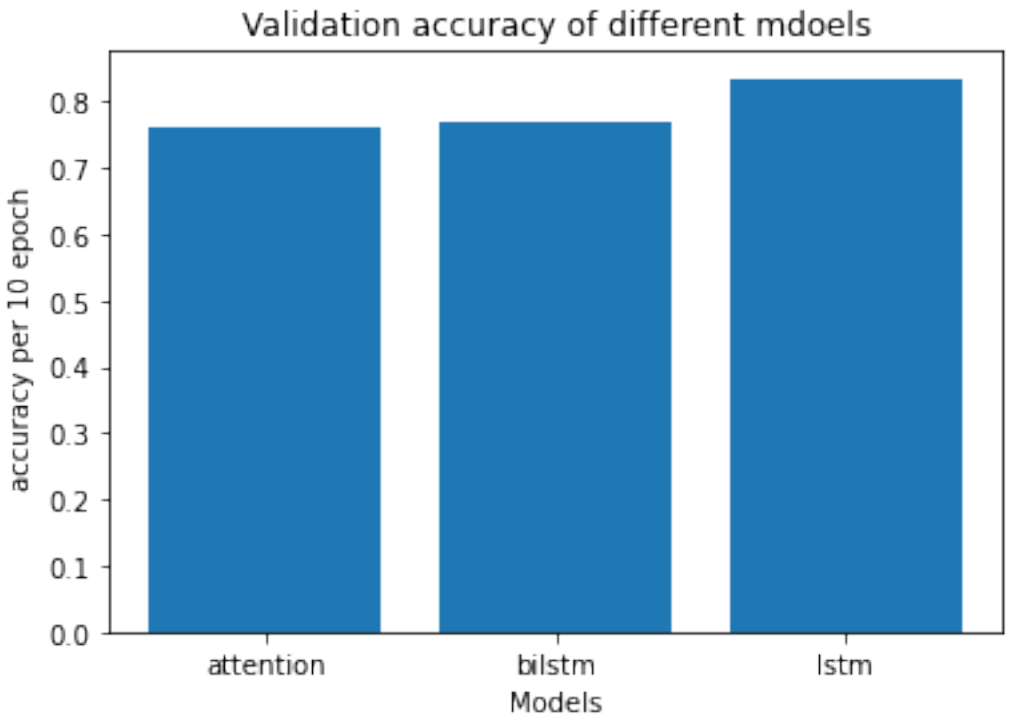
```
atten = max(model_atten.history.history['acc'])
bilstm = max(model_5.history.history['acc'])
lstm = max(model_3.history.history['acc'])
```

```
X = ['attention with lstm', 'bilstm' , 'lstm']
y = [atten, bilstm, lstm]

plt.bar(X,y)
plt.xlabel('Models')
plt.ylabel("accuracy per 10 epoch")
plt.title("Accuracy of Different Models")
plt.show()
```




```
atten = max(model_atten.history.history['val_acc'])
bilstm = max(model_5.history.history['val_acc'])
lstm = max(model_3.history.history['val_acc'])
X = ['attention', 'bilstm' , 'lstm']
y = [atten, bilstm, lstm]
plt.bar(X,y)
plt.xlabel('Models')
plt.ylabel(" accuracy per 10 epoch")
plt.title("Validation accuracy of different mdoels")
plt.show()
```


Validation accuracy of different mdoels
accuracy per 10 epoch
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
attention
bilstm
lstm
Models